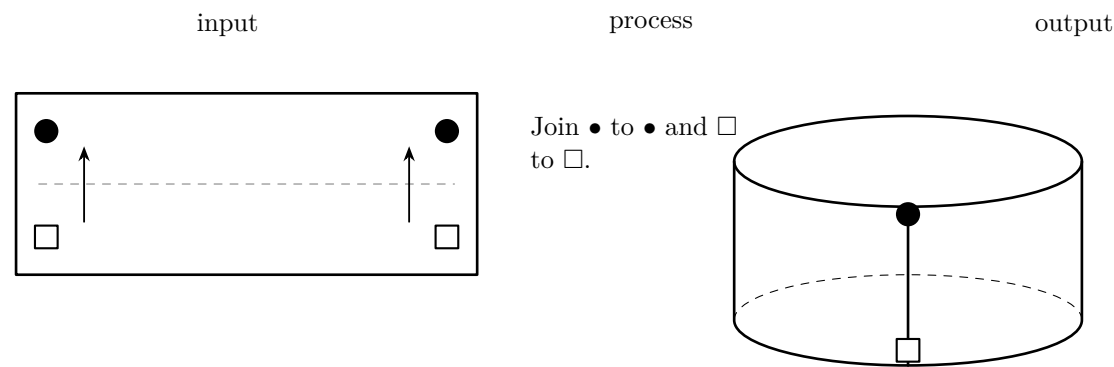
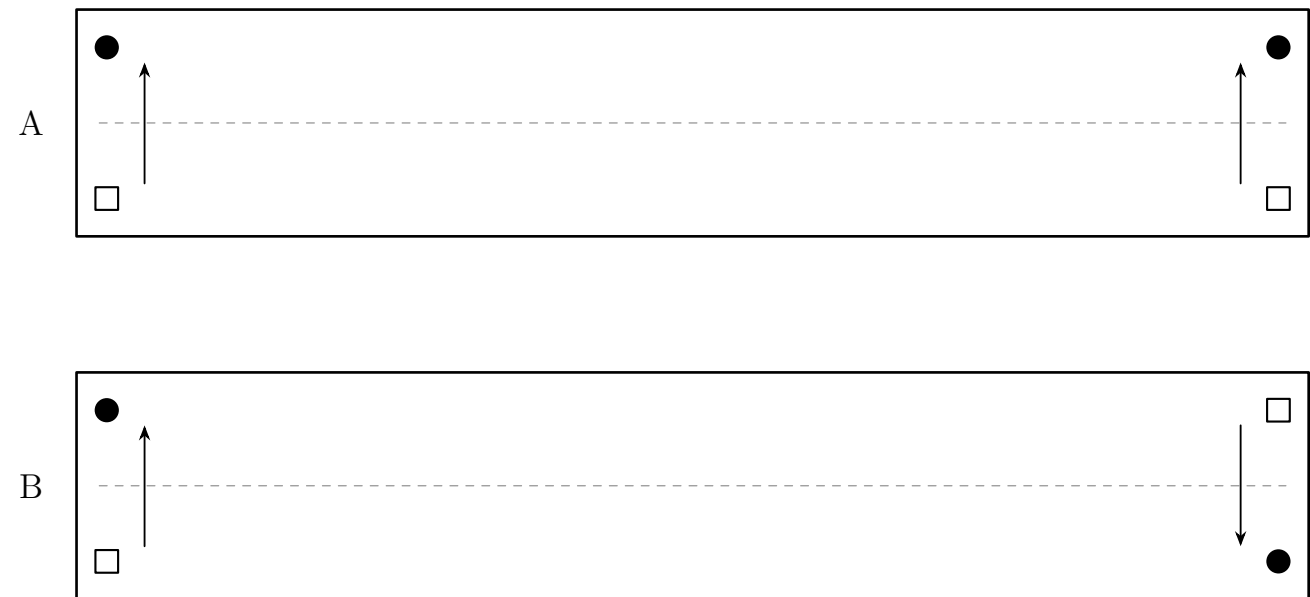


Keep the bands on the table. An adult makes every cut. The rectangles are joining recipes.



An edge trip stays on an edge, follows its seam marks, and stops on its first return to the start dot.

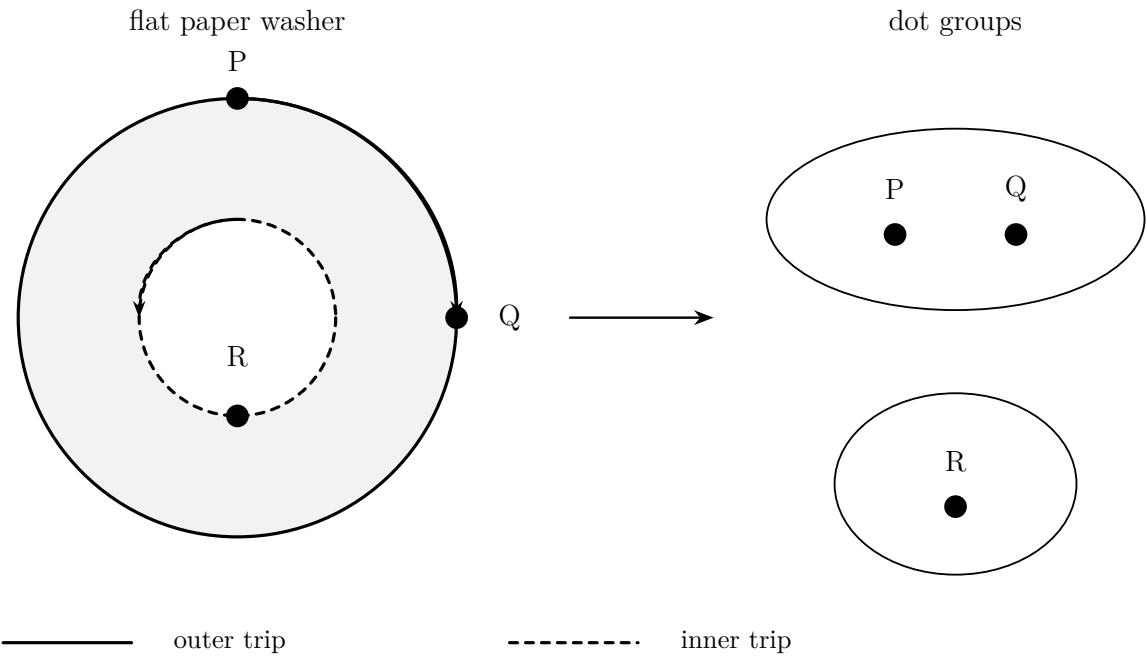
Problem 1: Trace all the edges of A and B. How many separate closed edges does each band have?



A: _____ B: _____

Problem 2: On A and B, find two edge dots that one edge trip can visit together, and two that it cannot. Which requests are impossible?

Dots on one edge go in one ring.

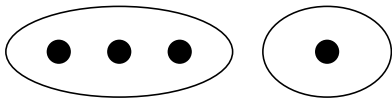


The rings record which dots share an edge.

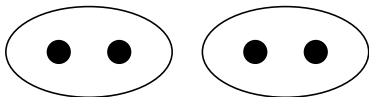
Problem 3: Place four edge dots to match each picture on A and on B, or cross it out. Each ring means a different closed edge; other edges may have no dots.



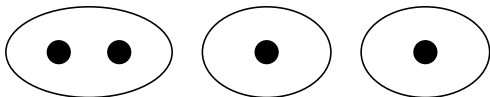
A: _____ B: _____



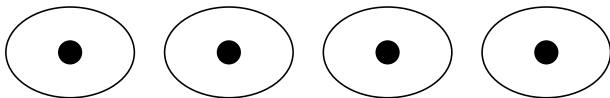
A: _____ B: _____



A: _____ B: _____

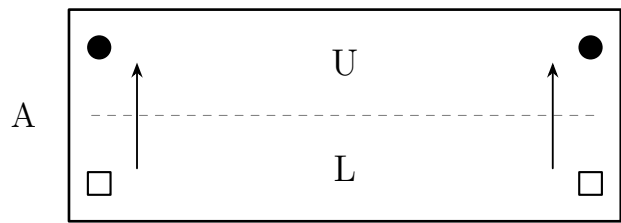


A: _____ B: _____



A: _____ B: _____

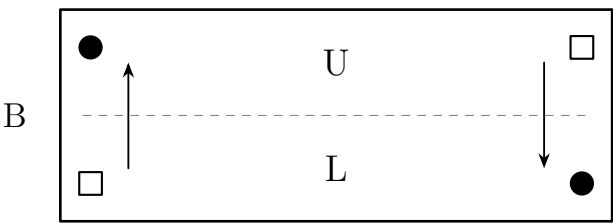
Problem 4: Predict how many connected pieces a middle-line cut will make in A and B. Ask an adult to make both cuts and record what happened.



I predict: _____



after cutting A

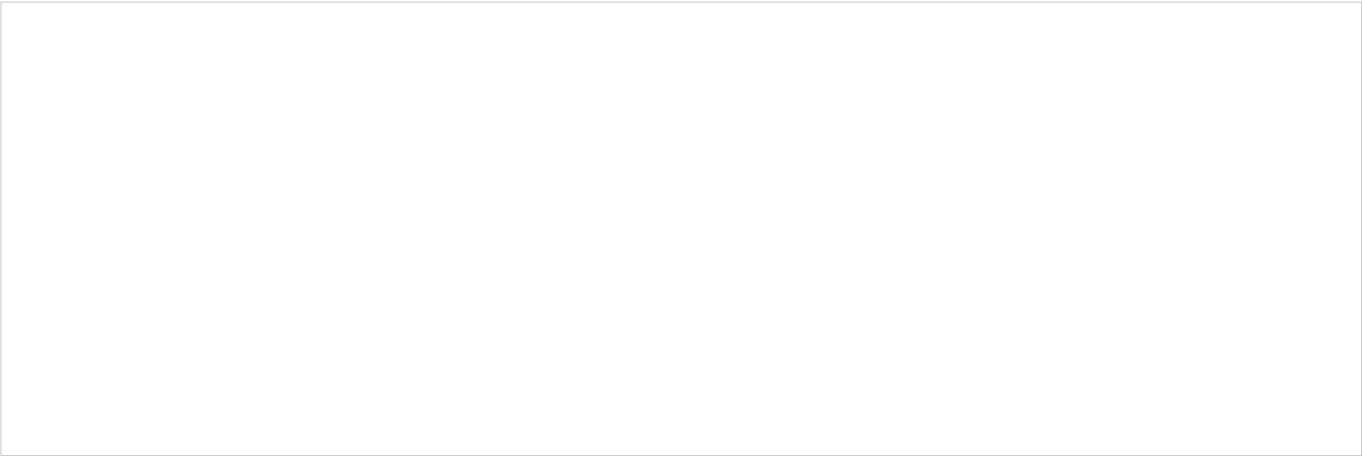
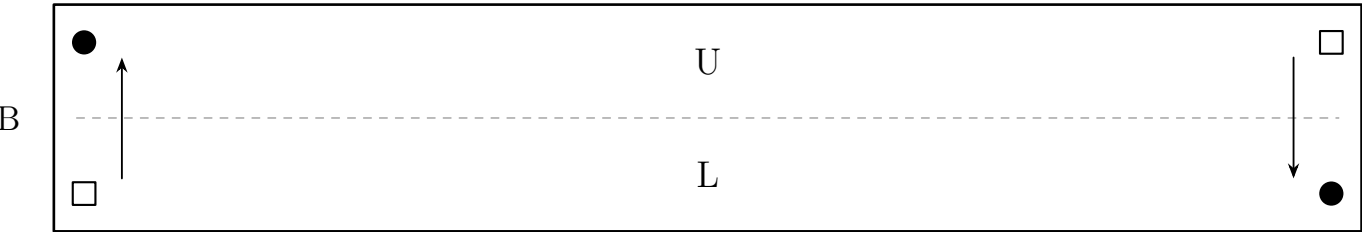
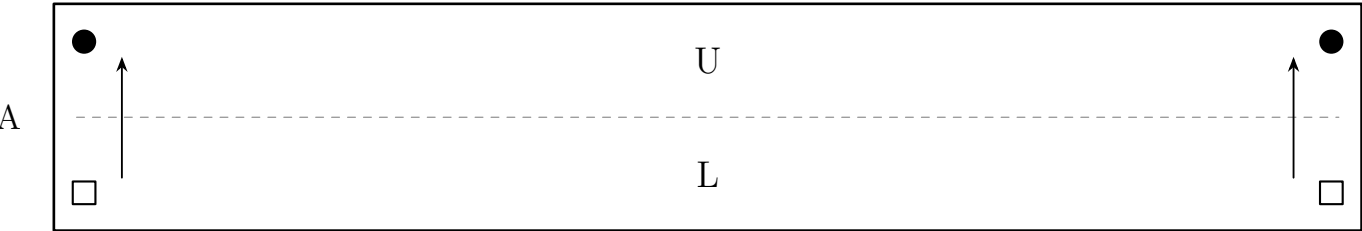


I predict: _____

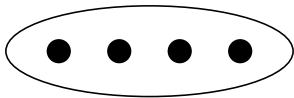


after cutting B

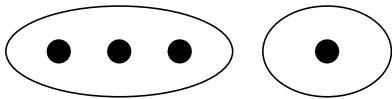
Problem 5: Use the U and L marks to explain why the cuts made that many pieces. On each cut result, can two edge dots lie on the same piece but on different closed edges?



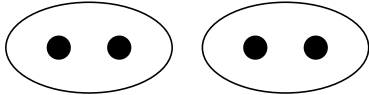
Problem 6: Use all the cut pieces from A together, then all those from B. Which four-dot pictures from Problem 3 can you now make?



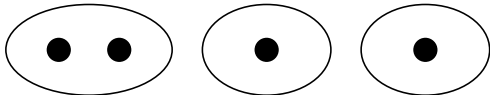
A: _____ B: _____



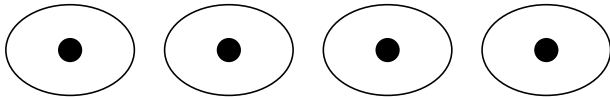
A: _____ B: _____



A: _____ B: _____



A: _____ B: _____



A: _____ B: _____

Problem 7: Arrange four edge dots so every cut piece has a dot and no edge trip reaches two dots. Can you do it with each cut result?